

Functional Decomposition of the Four Fundamental Interactions: A Minimal Physical Distinction-Operation Closure for Finite Distinction Systems

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The Standard Model and general relativity describe four known fundamental interactions: strong, electromagnetic, weak, and gravitational. X3 does not claim to derive the Standard Model gauge group, coupling constants, particle masses, scattering amplitudes, electroweak symmetry breaking, or quantum gravity. Instead, it proposes a functional decomposition of the four known interactions as a minimal physical distinction-operation closure for finite distinction systems. A finite physical system capable of maintaining persistent distinctions must implement four non-equivalent operation classes: hadronic/baryonic encapsulation of stable material tokens, connection and remote detectability among tokens, identity transformation or flavor-changing selective update, and global causal-geometric / stress-energy accounting. In the observed physical world these primary roles are realized respectively by the strong interaction, electromagnetism, the weak interaction, and gravity. The mapping is primary-role, not exclusive-role: real interactions can contribute to multiple physical processes, but each known interaction carries one operation class that the others do not generically replace. The paper separates the functional closure thesis from lower-status gauge-group minimality claims, distinguishes new particles or portals from genuinely new operation primitives, and gives falsification conditions for both. Deterministic normal-form demonstrations illustrate operation coverage, remove-one-operation failure, redundant fifth-mediator comparison, fifth-force classification, and the relation of X3 to nearby FDS physical-bridge papers.

Reader contract. X3 is a functional decomposition paper, not a Standard Model derivation paper. It does not derive $SU(3)_c \times SU(2)_L \times U(1)_Y$, coupling constants, masses, scattering amplitudes, CKM/PMNS parameters, electroweak symmetry breaking, or quantum gravity. It does not claim that no new particles, portals, dark-sector forces, fifth-force candidates, or effective interactions can exist. It does not identify gravity with dissipation. Its claim is narrower: the known four interactions naturally realize four non-equivalent physical operation classes required for finite distinction maintenance: encapsulation, connection, identity transformation, and global causal-geometric / stress-energy accounting.

Claim-status summary

Table I separates the functional closure claim from lower-status gauge-group and no-fifth-force interpretations.

Keywords

Finite Distinction Systems; Distinction Theory; fundamental interactions; strong interaction; electromagnetism; weak interaction; gravity; force taxonomy; operational closure; encapsulation; connection; identity transformation; flavor change; boundary maintenance; causal

geometry; stress-energy accounting; gauge theory; general relativity; fifth force.

INTRODUCTION

The force-taxonomy problem

Modern physics describes four known fundamental interactions. Quantum chromodynamics explains the strong interaction; electroweak theory explains the electromagnetic and weak interactions; general relativity describes gravity as spacetime geometry [7–11, 18]. This dynamical description is successful, but it leaves open a functional question: why do the known interactions occupy these four qualitatively different roles?

X3 does not ask whether FDS can derive the Standard Model. It asks a more modest and more operational question: why do the observed interactions naturally decompose into four non-equivalent operation classes needed by finite physical systems that maintain distinctions?

Central thesis

The thesis is that the four known interactions realize a minimal physical distinction-operation closure. The functional map is:

TABLE I. Central X3 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Finite distinction systems require token stabilization.	Operational / structural	Stable material tokens exist with no encapsulation, binding, exclusion, protection, or stabilization mechanism.
Strong interaction realizes hadronic/baryonic encapsulation.	Physical mapping	Hadronic stability is explained while confinement and color-neutral binding play no token-stabilizing role.
Finite distinction systems require remote detectability and compositional connection.	Operational / structural	Complex persistent structures form without any long-range or mediating interaction enabling detectable coupling.
Electromagnetism realizes connection and communication among charged sectors.	Physical mapping	Electromagnetism is shown not to underwrite chemistry, radiation, sensing, or long-range charged communication.
Finite distinction systems require identity transformation and selective update.	Operational / structural	Physical identity sectors never require conversion, decay, pruning, or irreversible update.
Weak interaction realizes identity transformation, flavor change, and unstable-state pruning.	Physical mapping	Flavor change, beta decay, and weak identity transformation are realized by another carrier.
Gravity realizes global boundary / causal geometry / stress-energy accounting.	Physical bridge	Gravity is shown to be unrelated to geometry, causal structure, horizons, stress-energy accounting, or global boundary constraints.
Four interactions form a minimal distinction-operation closure.	Main X3 thesis	A fundamental interaction implements a genuinely independent operation class not reducible to encapsulation, connection, identity transformation, or global boundary accounting.

Interaction	FDS primary role
Strong	hadronic/baryonic encapsulation stable tokens
Electromagnetic	connection / remote detectability communicable distinctions
Weak	identity transformation (flavor change) / pruning
Gravity	causal geometry stress-energy accounting

This map is not a gauge-group derivation. It is an operational decomposition.

Remark 1 (Primary-role, not exclusive-role). *The mapping is primary-role, not exclusive-role. Real interactions can contribute to multiple physical processes. Strong interactions scatter; electromagnetism stabilizes atoms and molecules; weak interactions propagate through gauge bosons; gravity shapes motion and energy flow. X3 classifies each interaction by the operation class it carries in a way the others do not generically replace.*

Why X3 follows X2

X2 analyzes a specific property of the weak row: a CP/T-oriented weak identity channel requires at least three CKM-participating weak flavors under the stated assumptions [6, 16, 17]. X3 asks why identity transformation is one of the necessary operation classes at all.

Thus X2 is a supporting module for the weak-interaction component of X3.

FDS BACKGROUND

Finite distinction systems

An active finite distinction system is represented by

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. X3 focuses on physical operation classes that stabilize X , connect Y to other distinctions, implement U when identity sectors change, and embed the system in geometric and resource constraints.

Physical distinction-operation requirements

A physical distinction is not only a label. It must be:

1. stabilized against perturbation;
2. connected to other distinctions and observation channels;

TABLE II. Operation-class independence in X3.

Operation	Cannot be replaced by
Encapsulation	connection without protected tokens
Connection	encapsulation without transmissible coupling
Identity transformation	ordinary scattering without sector conversion
Global boundary	local interaction without shared geometry

3. updated or transformed when its identity sector changes;
4. embedded in global causal, geometric, and stress-energy/resource constraints.

These four requirements motivate the operation basis

$$\mathcal{O}_{\text{FDS}} = \{\mathcal{O}_{\text{enc}}, \mathcal{O}_{\text{conn}}, \mathcal{O}_{\text{id}}, \mathcal{O}_{\text{geom}}\}, \quad (2)$$

where $\mathcal{O}_{\text{enc}}$ is encapsulation, $\mathcal{O}_{\text{conn}}$ is connection / remote detectability, $\mathcal{O}_{\text{id}}$ is identity transformation / selective update, and $\mathcal{O}_{\text{geom}}$ is global boundary / causal geometry / stress-energy accounting.

Definition 1 (Distinction-operation closure). *A physical operation set is closed for finite distinction maintenance when it provides enough mechanisms to stabilize tokens, connect tokens, transform identity sectors, and embed all local operations in global causal and resource constraints.*

Remark 2 (Closure, not derivation). *Equation (2) is a functional closure basis. It is not a derivation of $SU(3)_c \times SU(2)_L \times U(1)_Y$ or of the Einstein field equations. It states what kinds of operations must be physically instantiated if stable distinction-maintaining systems exist.*

Operation-class independence

Two physical mechanisms may be dynamically coupled while still implementing distinct operation classes. X3 classifies by irreducible function under a fixed accounting boundary, not by isolated force ontology. Let F_j denote the four required functions: token stabilization, remote detectability, identity-sector conversion, and causal-geometric accounting. An operation class $\mathcal{O}_k$ is independent when removing it leaves at least one required function under-implemented by the remaining classes without changing the definitions of the remaining classes or expanding the accounting boundary.

This independence criterion is functional rather than metaphysical. It allows cross-coupling between interactions while preserving the claim that each operation class supplies one primary capability the others do not generically provide.

OPERATION I: STRONG INTERACTION AS ENCAPSULATION

Functional need

A finite physical system requires stable material tokens. Without token stabilization, persistent addressable states, composite structures, and memory-bearing substrates do not exist. In X3, encapsulation means binding raw degrees of freedom into protected tokens.

Physical realization

The strong interaction realizes the bottom hadronic/baryonic token-stabilization layer through color confinement, color-neutral hadron formation, and nuclear binding. QCD is not replaced by FDS; X3 interprets confinement and color-neutral binding as the observed physical implementation of hadronic token encapsulation [7, 12–14].

Strong encapsulation is not the sole stabilization mechanism for all composite structures. Atomic, molecular, and solid-state stability depend primarily on electromagnetic and quantum-statistical structure. The X3 claim is narrower: QCD supplies the lowest hadronic/baryonic material-token layer on which ordinary matter is built.

Gauge-group caveat

The observed implementation is $SU(3)_c$. X3 does not uniquely derive $SU(3)$ over $SO(N)$, $Sp(N)$, G_2 , or other alternatives. It treats $SU(3)_c$ as the Standard Model realization of the encapsulation role. A rigorous no-go theorem excluding alternative confining structures is a separate higher-risk problem.

Failure if absent

Without encapsulation there are no stable hadrons, no stable nuclei, and no persistent baryonic matter tokens. A world with connection and geometry but no encapsulated matter substrate lacks the material carriers required for long-lived distinctions.

OPERATION II: ELECTROMAGNETISM AS CONNECTION

Functional need

Encapsulated tokens must interact, compose, and become detectable. A universe of perfectly isolated tokens could store local distinctions but could not form atoms,

molecules, sensors, or communication channels. Connection means making distinctions mutually detectable and composable.

Physical realization

Electromagnetism realizes connection through long-range interactions among charged sectors, atomic and molecular binding, radiation, sensing, and photon-mediated communication. Electromagnetism is not merely a force between charges; it is the dominant channel by which ordinary matter becomes chemically structured, optically detectable, measurable, and communicable at macroscopic scales [7, 15].

Long-range detectability

The unbroken $U(1)_{\text{EM}}$ and masslessness of the photon provide the observed long-range connection channel. X3 does not derive photon masslessness. It interprets unbroken electromagnetism as the observed physical channel that permits distinctions to remain remotely detectable across macroscopic scales.

Failure if absent

Without electromagnetism, there are no ordinary atoms, chemical bonds, photons, optical sensing, radio communication, or electrically mediated macroscopic organization. Encapsulation without connection yields inert protected tokens rather than complex interacting structures.

OPERATION III: WEAK INTERACTION AS IDENTITY TRANSFORMATION

Functional need

Finite distinction systems cannot merely preserve identities. They must also allow identity-sector transitions: flavor conversion, decay of unstable configurations, selective pruning, and irreversible state update. In X3, identity transformation means controlled change of particle type, flavor sector, or decay channel. The weak interaction's chirality and CP/T violation are physical signatures that this identity update has a built-in directional character: if identity transformation were fully reversible and non-dissipative, it would belong to the connection class (operation II) rather than constituting a distinct pruning-like operation.

Here identity transformation means flavor, particle-sector, or decay-channel conversion, not arbitrary state

evolution, scattering, excitation, or binding rearrangement. Strong and electromagnetic interactions certainly change physical states; the weak role singled out in X3 is change of identity sector.

Physical realization

The weak interaction realizes identity transformation through beta decay, quark flavor change, neutrino interactions, chirality-sensitive charged currents, and CP/T-oriented weak-sector transformation. A representative quark identity transformation is

$$\mathcal{T}_{\text{id}} : |d_j\rangle \mapsto \sum_i V_{ij} |u_i\rangle, \quad (3)$$

where V_{ij} is the CKM matrix [6, 16, 17].

Pruning and capacity overflow

In FDS language, weak-sector identity transformation is the observed microscopic carrier of a pruning-like operation: not arbitrary deletion, but selective transition of unstable identity sectors into more stable or lower-energy configurations. Without such a channel, unstable sectors would either persist without decay or require another mechanism for clearing identity configurations that cannot remain encapsulated. This connects weak decay to the FDS problem of capacity overflow without claiming that all pruning in all systems is weak interaction.

Why selective weakness matters

If identity transformation were uncontrolled or too rapid relative to encapsulation and connection, stable structures would be continuously destabilized. The empirical weakness and mass gap of weak interactions are consistent with the requirement that identity update be selective rather than universal. This is a functional consistency claim, not a derivation of M_W , M_Z , or the Fermi constant.

Failure if absent

Without weak identity transformation, there is no beta decay, no observed quark flavor transitions, no ordinary weak nuclear reaction chain as observed in stars, and no CP/T-oriented weak identity channel. Stable tokens could exist and connect, but identity sectors could not be selectively updated.

OPERATION IV: GRAVITY AS GLOBAL BOUNDARY, CAUSAL GEOMETRY, AND STRESS-ENERGY ACCOUNTING

Functional need

A finite distinction system is embedded in causal structure. Distinctions require global boundary conditions, causal ordering, horizon structure, energy-momentum accounting, geometric response to stress-energy, and constraints on localization and information flow.

Physical realization

Gravity realizes these roles through spacetime geometry, causal cones, horizon structure, curvature sourced by stress-energy, and universal coupling to energy-momentum [18–21]. In X3, gravity is the observed physical implementation of the global boundary / causal geometry / stress-energy accounting class. In FDS terms, curvature encodes a wiring or addressing cost: as the density of maintained distinctions (mass-energy) increases, the local geometric constraints on communication and causal ordering tighten, which is the general-relativistic manifestation of finite-system boundary pressure.

Gravity is not merely dissipation

X3 does not identify gravity with dissipation. Dissipation and resource accounting are ledger channels inside gravitational boundary dynamics. Gravity itself is the independent geometric field governing causal structure and global boundary conditions. Gravitational waves, horizons, and spacetime curvature are not reduced to heat production. Stress-energy accounting is the physics-side language; resource accounting is the FDS interpretation of why global boundary constraints matter. When erasure or reset operations are physically implemented, Landauer’s principle supplies a baseline thermodynamic reference for the resource ledger, but gravity as a geometric sector is not itself a dissipative channel [28].

Equivalence-principle caveat

The equivalence principle may admit an FDS interpretation as a consistency condition between inertial book-keeping and gravitational boundary response, but this is not used in X3. It remains future work rather than part of the closure argument.

Failure if absent

Without gravity there is no dynamical spacetime geometry, no universal stress-energy accounting, no horizon-scale boundary structure, and no global causal constraint tying local operations into a common physical world.

MINIMAL DISTINCTION-OPERATION CLOSURE

Joint sufficiency and individual necessity

The four classes jointly provide stable tokens, interactions among tokens, transformation of tokens, and embedding in causal/geometric boundary structure. Remove one class and the closure fails:

Removed operation	Consequence
Encapsulation	no stable material tokens
Connection	no remote detectability or chemistry
Identity transformation	no flavor conversion, decay, or pruning
Global boundary	no causal geometry or horizon accounting

Proposition 1 (Functional distinction-operation closure). *For a physical finite distinction system with stable material tokens, remote detectability, identity-changing dynamics, and global causal geometry, at least four non-equivalent operation classes are required: $\mathcal{O}_{\text{enc}}$, $\mathcal{O}_{\text{conn}}$, $\mathcal{O}_{\text{id}}$, and $\mathcal{O}_{\text{geom}}$. In the observed physical world, the strong, electromagnetic, weak, and gravitational interactions realize these operation classes as primary roles.*

Proof. By construction, stability of material tokens requires an encapsulation/protection operation. Detectability and composition require a connection operation. Flavor or particle-sector conversion, decay, and selective update require an identity-transformation operation. Embedding local operations in a consistent causal and stress-energy-constrained world requires a global boundary/geometric operation. By the independence criterion in Table II, removing any one class leaves one required function under-implemented by the remaining classes. This proves a functional closure proposition, not a unique gauge-dynamics derivation. $\square$

FIFTH-FORCE AND NEW-PHYSICS CAVEAT

New particle versus new operation class

X3 does not rule out new particles, portals, sectors, or effective interactions. A dark photon, scalar mediator, axion-like particle, hidden confining gauge force, or

long-range spin-dependent interaction may extend physical phenomenology without adding a new distinction-operation primitive [25–27, 29–31]. A hidden confining force may be encapsulation-like; a dark photon may be connection-like; a scalar modifying masses may be boundary/resource-like; a new decay channel may be identity-update-like. Recent reviews of exotic spin-dependent interactions and asteroid-tracking constraints on fifth forces illustrate that many candidates are mediated by new bosons or are searched for as precision deviations, but such signals would challenge X3 only if they implement a new operation primitive [29, 30].

Falsification of closure

X3 is challenged only by a fundamental interaction implementing a new irreducible operation class not reducible to encapsulation, connection, identity transformation, or global boundary accounting. In that case, $\mathcal{O}_{\text{FDS}}$ must be expanded.

Criterion 1 (Fifth-operation criterion). *A candidate new force falsifies or expands X3 only if it implements a physically necessary operation primitive that cannot be classified as an instance, sector extension, or mixture of $\mathcal{O}_{\text{enc}}$, $\mathcal{O}_{\text{conn}}$, $\mathcal{O}_{\text{id}}$, or $\mathcal{O}_{\text{geom}}$.*

GAUGE GROUP DISCUSSION: LOWER-STATUS MAPPING

The observed structures map to the operation classes as follows:

Operation	Observed implementation
Encapsulation	$SU(3)_c$ QCD
Identity transformation	weak charged-current / broken electroweak sector
Long-range connection	unbroken $U(1)_{\text{EM}}$
Global boundary	diffeomorphism invariance / GR

This mapping motivates why the observed interactions have different mathematical forms. It does not uniquely derive the groups. X3 does not exclude all alternative gauge groups; does not derive electroweak symmetry breaking; does not calculate coupling constants; and does not quantize gravity. These remain lower-status problems.

Higgs mechanism as implementation layer

The Higgs mechanism is not treated as a fifth operation class in X3. It parametrizes mass generation and electroweak symmetry breaking inside the implementation of encapsulation, identity transformation, and long-range connection [7, 22–24]. In FDS language, masses

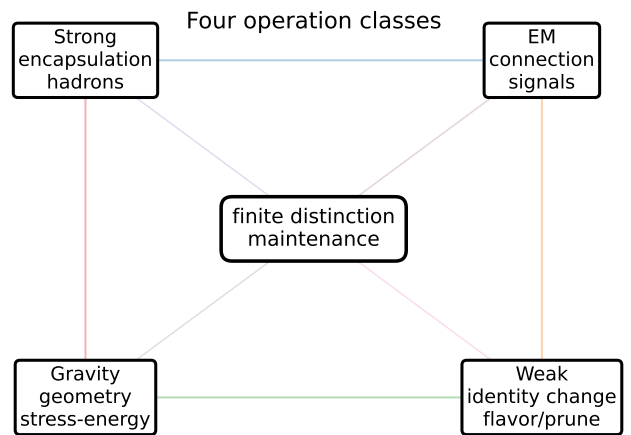


FIG. 1. Four known interactions as operation classes for finite distinction maintenance. The center represents finite distinction maintenance; the four nodes represent non-equivalent physical operation classes realized by the known interactions.

may be interpreted as energetic separation scales between distinction sectors, but X3 does not derive the Higgs potential, vacuum expectation value, or Yukawa texture.

Remark 3 (Gauge-group caveat). *Claims about $SU(3)$, $SU(2)$, $U(1)$, or diffeomorphism invariance are implementation mappings. If another group structure realizes the same operation role in a consistent physical theory, X3 should demote the corresponding gauge-specific claim while retaining the functional role.*

NORMAL-FORM DEMONSTRATIONS

The figures are deterministic normal-form demonstrations generated by `code/generate_results.py`. They are not empirical fits and do not simulate Standard Model dynamics. The normal-form model defines a qualitative coverage matrix between operation classes and requirements, enumerates operation subsets, and computes residual closure loss. Matrix entries are primary-role scores, not coupling constants, fitted data, or empirical weights. The only intended claim is conceptual: removing any one operation class leaves a distinct functional deficit, while adding a redundant fifth mediator without a new operation primitive does not change closure.

RELATION TO NEARBY FDS PAPERS AND EXISTING PHYSICS

X3 is compatible with QCD, electroweak theory, CKM/PMNS flavor physics, general relativity, and quantum field theory. It is not a replacement for them. It packages a functional implication inside FDS: per-

TABLE III. Relation of X3 to nearby FDS papers and established theories.

Theory / paper	What it provides	X3 use
QCD	Confinement, color dynamics, hadrons, nuclear matter	Observed implementation of hadronic/baryonic encapsulation.
QED / electromagnetism	Long-range photon-mediated interaction, atoms, radiation, sensing	Observed implementation of connection / remote detectability.
Electroweak theory	Chiral weak currents, beta decay, flavor change, EWSB	Observed implementation of identity transformation.
General relativity	Causal geometry, curvature, horizons, stress-energy accounting	Observed implementation of global boundary / geometry.
P3/P4/P6/P7	Environmental records, internal forgetting, throughput bounds, protected invariants	Nearby FDS physical-bridge constraints on boundary maintenance [2–5].
X2	CKM lower bound for CP/T-oriented weak identity transformation	Weak-interaction module inside the X3 taxonomy.
Fifth-force searches	New particles, portals, deviations, hidden-sector mediators	Classification test: new mediator or new operation primitive?

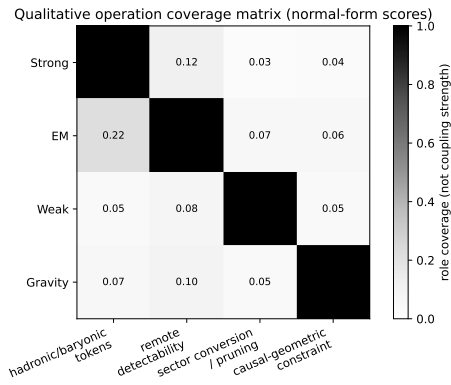


FIG. 2. Qualitative coverage matrix. The entries are normal-form scores, not coupling constants, fitted data, or empirical weights. Diagonal dominance expresses primary-role coverage; weak off-diagonal entries mark cross-role participation in real physical systems. Scores are schematic; they represent a logarithmic proxy for the relative dominance of each interaction in its primary role, not fitted coupling magnitudes.

sistent distinctions require stabilization, connection, update, and global boundary accounting.

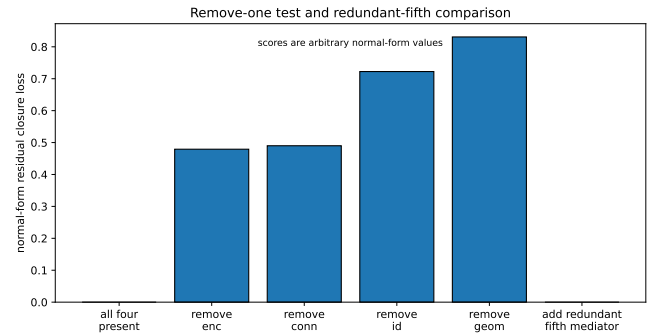


FIG. 3. Remove-one-operation test and redundant-fifth comparison. Absolute values are arbitrary normal-form scores; only the nonzero residual after removing each operation class is intended. A redundant fifth mediator that adds no new operation primitive does not improve closure.

Fifth-force audit: new mediator vs. new operation primitive

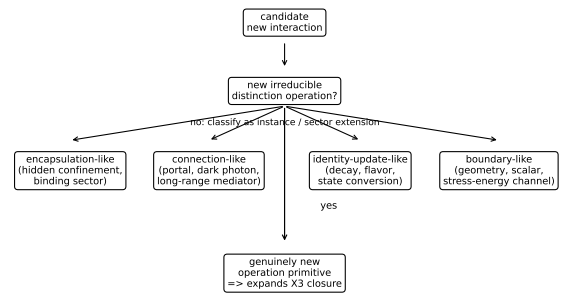
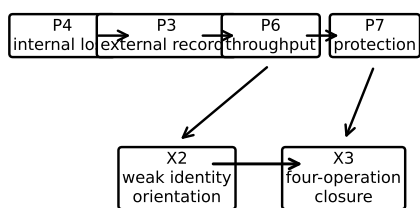


FIG. 4. Fifth-force audit. X3 distinguishes new particles or sector-specific mediators from a genuinely new operation primitive. Only the latter expands the X3 closure basis.

X3 relation map



X2 supplies the weak-sector module; X3 places it in the operation taxonomy.

FIG. 5. Relation of X3 to nearby FDS physical-bridge papers. X2 supplies a weak-sector identity-transformation module; X3 uses it inside a four-operation taxonomy.

DESIGN IMPLICATION FOR PHYSICAL AI

This section is not part of the physics proof. It extracts the FDS design rule implied by X3. A boundary-maintaining agent requires the same four functional classes: stable tokens, communication links, identity update, and global resource/boundary accounting.

Missing operation class	Agent failure mode
Encapsulation	identity drift / unstable object tokens
Connection	isolated modules / broken relation graph
Identity transformation	stale state / dead logic / overfitting
Global boundary	resource overshoot / unsafe action

The lesson is not that agents need copies of physical forces. It is that stable boundary maintenance requires a minimal operation closure analogous to the physical one: protected tokens, communication, identity revision, and global resource / safety-envelope accounting.

FALSIFICATION AND DEMOTION CONDITIONS

X3 is weakened if stable physical tokens exist without any encapsulation/protection operation; if complex structure and observation exist without connection/detectability; if identity-sector transitions exist without transformation/update; or if global causal geometry / stress-energy accounting is irrelevant to physical distinction maintenance.

X3 is falsified or expanded if a fundamental interaction is found that implements a necessary operation class not reducible to encapsulation, connection, identity transformation, or global boundary accounting. Gauge-specific claims are lower-status: if alternative group structures realize the same operation roles, the gauge mapping should be demoted rather than the closure thesis.

CONCLUSION

X3 proposes that the four known interactions occupy four non-equivalent functional roles required by finite distinction systems. The strong interaction stabilizes hadronic/baryonic material tokens; electromagnetism connects and exposes distinctions to remote detection; the weak interaction transforms identity sectors and enables selective flavor/particle-sector update; gravity supplies global causal geometry and stress-energy/boundary accounting. This is not a derivation of the Standard Model, nor a claim that no new particles or sector-specific forces exist. It is a closure claim: any physical world supporting persistent distinctions must implement these op-

eration classes, and the known interactions realize them in our universe.

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